WEIJIE, WANG

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Education

Geometry Dynamics Co., Ltd. — Dec 2025 – Now
Motion Control Algorithm Lead — *Beijing, China*

Italian Institute of Technology — Nov 2023 – Nov 2025
Researcher — *Genova, Italy*

University of Chinese Academy of Sciences — Sep 2020 – July 2023
Master of Electronic Information — *Beijing, China*

- Key Courses: Advanced Engineering Mathematics, Intelligent Mobile Robots,Introduction to Algorithms, Pattern Recognition and Machine Learning, Computer Vision: Algorithms and Applications

Henan University of Technology — Sep 2015 – Jun 2019
Bachelor of Automation — *Henan, China*

- Key Courses: Automatic Control Theory, Modern Control Theory, Computer Control Technology, Process Control Systems, Motion Control System, Functions of Complex Variables Integral Transformations, Circuit, Analog Electronics Technology, Digital Electronics Technology, Microcontroller Principle and Interface technology, Robotics Measurement and Control Technology

Projects

Wheeled-Legged Last-Mile Delivery Robot — Jun 2025 – Jul 2025
Geometry Dynamics Co., Ltd. — *Beijing, China*

- **Autonomous last-mile delivery.** A wheeled-legged mobile robot for autonomous last-mile delivery—pairing wheeled efficiency on flat ground with legged mobility to negotiate curbs, steps, and mixed indoor/outdoor terrain, carrying parcels the final stretch to the door. I develop the whole-body motion-control algorithms behind its stable locomotion and mode transitions.

Vision-Based Quadruped Loco-Manipulation — Jan 2025 – Apr 2025
Geometry Dynamics Co., Ltd. — *Beijing, China*

- **Multi-functional operation across diverse terrains.** Vision-driven loco-manipulation for a quadruped robot, coupling onboard perception with whole-body control to perform versatile manipulation and locomotion tasks across diverse, multi-terrain environments.

General-Purpose Humanoid Robot Platform — Dec 2025 – Mar 2025
Geometry Dynamics Co., Ltd. — *Beijing, China*

- **Whole-body motion control & dynamic loco-manipulation.** A full-size, general-purpose humanoid robot developed as a turnkey ODM platform. I lead the whole-body motion-control stack—real-time balancing, compliant whole-body control, and dynamic loco-manipulation—fusing learning-based motor skills with model-based trajectory optimization across a high-degree-of-freedom body. The system achieves robust bipedal locomotion and precise bimanual, multi-finger manipulation in unstructured environments, and is engineered for rapid customization across industrial and service scenarios.

Research Experience

INTENTION: Humanoid Robot Motion Intention via VLM — Oct, 2024 – Apr, 2025
Italian Institute of Technology — *Genova, Italy*

- Traditional control and planning for robotic manipulation heavily rely on precise physical models and predefined action sequences. While effective in structured environments, such approaches often fail in real-world scenarios due to modeling inaccuracies and struggle to generalize to novel tasks. In contrast, humans intuitively interact with their surroundings, demonstrating remarkable adaptability, making efficient decisions through implicit physical understanding. In this work, we propose INTENTION, a novel framework enabling robots with learned interactive intuition and autonomous manipulation in diverse scenarios, by integrating Vision-Language Models (VLMs) based scene reasoning with interaction-driven memory. We introduce Memory Graph to record scenes from previous task interactions which embodies human-like understanding and decision-making about different tasks in real world. Meanwhile, we design an Intuitive Perceptor that extracts physical relations and affordances from visual scenes. Together, these components empower robots to infer appropriate interaction behaviors in new scenes without relying on repetitive instructions.
- Papers: INTENTION: Inferring Tendencies of Humanoid Robot Motion Through Interactive Intuition and Grounded VLM (Humanoids)

Learning Hybrid Behavior Planning for Autonomous Loco-manipulation Nov, 2023 – Oct, 2024
Italian Institute of Technology *Genova, Italy*

- Enabling robots to autonomously perform hybrid motions in diverse environments can be beneficial for long-horizon tasks such as material handling, household chores, and work assistance. This requires extensive exploitation of intrinsic motion capabilities, extraction of affordances from rich environmental in- formation, and planning of physical interaction behaviors. Despite recent progress has demonstrated impressive humanoid whole-body control abilities, they strug- gle to achieve versatility and adaptability for new tasks. In this work, we propose HYPERmotion, a framework that learns, selects and plans behaviors based on tasks in different scenarios. We combine reinforcement learning with whole-body optimization to generate motion for 38 actuated joints and create a motion library to store the learned skills. We apply the planning and reasoning features of the large language models (LLMs) to complex loco-manipulation tasks, constructing a hierarchical task graph that comprises a series of primitive behaviors to bridge lower-level execution with higher-level planning. By leveraging the interaction of distilled spatial geometry and 2D observation with a visual language model (VLM) to ground knowledge into a robotic morphology selector to choose appro- priate actions in single- or dual-arm, legged or wheeled locomotion. Experiments in simulation and real-world show that learned motions can efficiently adapt to new tasks, demonstrating high autonomy from free-text commands in unstruc- tured scenes.
- Papers: HYPERmotion: Learning Hybrid Behavior Planning for Autonomous Loco-manipulation(The Conference on Robot Learning (CoRL))

Highly dynamic locomotion control of biped robot with swing arms Jan, 2021 – July, 2023
Institute of Automation, Chinese Academy of Sciences *Beijing, China*

- Jumping as the basis for highly dynamic locomotion of biped robots, biomechanical research results show that the introduction of arm-swing during jumping can effectively improve the stability and agility of locomotion and reduce energy consumption. I present a kind of control strategy that integrates swing arms into the whole realization process of the jumping motion of biped robot: modeling, trajectory planning and tracking, which models the biped robot as a on Flywheel Dynamics Spring Model (FDSM) to extract characteristics of swing arms and uses the whole-body controller (WBC) to achieve them. Simulation of jumping motion is implemented on a kind of biped robot Purple V1.0 designed for high explosive locomotion, and an evaluation method is proposed which includes three aspects of agility, stability and energy consumption. Experiment results show the control strategy's effectiveness and agility is increased by a maximum of 6%, stability is enhanced by an average of 30%, and energy consumption is reduced by an average of 15%
- Papers: Highly dynamic locomotion control of biped robot with swing arms, Accepted(IEEE International Conference on Advanced Robotics and Mechatronics)

Research on safe autonomous motion planning for mobile robots Jun, 2021 – Dec, 2021
Institute of Automation, Chinese Academy of Sciences *Beijing, China*

- The safe and efficient autonomous motion planning capability is a prerequisite for mobile robots to adapt to complex and unknown environments. To achieve safe and efficient autonomous motion and high passage rate of robots, I proposed a safe and efficient autonomous motion planning method based on forward reachable set (offline planning) and deep reinforcement learning (online decision making), and conduct comprehensive experiments on the physical simulation engine RoboMaster, and the results show that the method achieves safe autonomous motion of robots and Compared to traditional methods, this method achieves safe autonomous robot movement and a 30 % increase in the rate of passing complex environments
- Papers: Improving the Ability of Robots to Navigate Through Crowded Environments Safely using Deep Reinforcement Learning, Accepted(IEEE International Conference on Advanced Robotics and Mechatronics)

Multi-mobile robot task assignment and scheduling system design Aug, 2020 – May, 2021

Institute of Automation, Chinese Academy of Sciences *Beijing, China*

- Based on the SLAM mapping, pose estimation, path planning and tracking control, the multi-robot task assignment and optimal path scheduling with minimum resource utilization are implemented in the Matlab simulation environment based on the collision detection search algorithm and the multi-intelligent task assignment algorithm, and are experimentally verified in the test site based on ROS with good results. On the basis of this, we consider the lifelong working mode, the collision On the basis of this, the simulation is validated by considering the lifelong working mode, extending the collision set, using dynamic planning to increase and decrease the robots online, and ensuring the optimal planning.